

7.4 AI Behavioural Screening Agent

Objective: Analyse the applicant's short video interview across five independent, purpose-built signals, each targeting one specific, narrow question. This agent does not produce a single "credibility" or deception verdict. A poor result on any component is shown only as an advisory flag for the Branch Manager / Loan Approval Committee to review – it is never subtracted from any score. The composite result across the four scored components contributes continuously to the final Trust Score (Section 8.7), never to the Credit Score (Section 8.4). Each component uses a purpose-built, validated tool for its specific task – gaze tracking, secondary-voice detection, and liveness/anti-spoofing are each established, separate technical disciplines, and are treated as such here.

Features:

- In-app video capture, 30-90 seconds; applicant states name, loan purpose, and answers basic questions (finalised with Akhuwat team).
- Five independent analysis components (below), each producing its own output.
- Audio extracted separately from video for the Answer Verification component.
- Back-office portal shows all five component outputs side-by-side, with timestamps for any flagged moments.
- Video stored encrypted, accessible only to authorised Loan Committee members, deleted 7 days post-decision.

The Five Components:

Component	What it measures	Weight
1. Gaze Stability	Tracks eye/iris position across the video; % of time gaze remained on camera. Proxy for scripted-reading or off-camera coaching.	40% of composite
2. Answer Verification	Audio transcribed, compared against the applicant's submitted application data (name, loan purpose) for factual consistency. Flags factual mismatches only – never tone, sentiment, or perceived truthfulness.	35% of composite
3. Acoustic Environment Check	Detects secondary voices / overlapping	15% of composite

	speech in the audio. Reports speaker count and overlap timestamps.	
4. Background Anti-Spoofing	Confirms a genuine live face is present vs. photo/screen replay/mask. Cross-verified against CNIC photo on file.	10% of composite
5. Vocal Stress Indicators	Measures jitter, shimmer, pitch perturbation – physical vocal strain markers. Reports timestamps of stress spikes only. Contributes no numeric value to any score, in either direction.	Advisory flag only – not scored

Facial emotion / micro-expression analysis is deliberately not used by this agent. Facial expression norms vary significantly across cultures, and available models have no validated performance data for Akhuwat's applicant population. This kind of analysis also requires the system to judge what emotion an applicant "should" display for a given topic, then flag deviation – a normative judgement fundamentally different from, and less defensible than, a physical measurement like gaze or vocal jitter. A first-time applicant unfamiliar with being on camera may display flat or restrained affect while telling the complete truth, and the system cannot reliably distinguish this from any other cause of the same expression.

Key Requirements & Business Rules:

- Minimum video duration: 30 seconds; Maximum: 90 seconds.
- Applicant must clearly state name and loan purpose; full face visible throughout, no obstructions.
- Poor lighting or audio quality triggers automatic re-recording request.
- Format: MP4; Resolution: minimum 720p; Frame rate: 24-30fps; Size: <= 50MB.
- Videos encrypted and deleted after 7 days post-decision.
- Mandatory video for all loan amounts.
- Computation must complete within 30 seconds of application submission.
- The Behavioural Consistency Signal (composite of components 1-4) is dual-direction: a poor result is advisory only. A clean/consistent result across all four components applies a small positive bonus to the Trust Score. Vocal Stress (component 5) never affects any score in either direction.
- NADRA/NACTA/UNSC/blacklist results (Section 7.1) are separate hard-stops and are not part of this agent's dual-direction logic.

Behavioural Consistency Signal – Composite Calculation:

Behavioural_Consistency_Signal (0-100) = (Gaze_Stability x 0.40) + (Answer_Verification x 0.35) + (Acoustic_Check x 0.15) + (Background_Antispoofing x 0.10)

This composite score contributes continuously to the final Trust Score, computed in Section 8.7 (Behavioural Score / 100 x 80 points, capped at 80). A higher score contributes more, a lower score contributes less; there is no all-or-nothing threshold. This composite is never subtracted from, and never contributes to, the Credit Score (Section 8.4) itself; it only feeds the final Trust Score assembled in Section 8.7.

Continuous Improvement:

- Rule-based components (Gaze, Acoustic, Background): accuracy tracked against manager agreement/override rate over time; retrained on the same quarterly cadence as Agent 8.4 and 8.9.
- LLM-reasoning component (Answer Verification): model/prompt version changes tracked as versioned events. Compare agreement rate with manager review outcomes before fully switching versions.
- Per Section 9.3a, every application's full behavioural output is stored, whether or not a human ever reviewed or overrode it.

User Expectations:

- As a borrower, I want clear instructions for video submission.
- As an applicant, I expect privacy assurances and quick feedback.

Visibility:

In mobile app, a dedicated video record screen with preview and guidelines. Back-office portal displays all five component outputs side-by-side with timestamps for flagged moments.

Acceptance Criteria:

- Screening completes in <30 seconds for videos.
- Flagged applications trigger notifications instantly.
- Upload sample video with inconsistencies; system flags each of the 5 components correctly and independently.

